



Insurance Loss Analytics : A Prediction & Classification Problem

Problem Statement

- **Problem**
 - Predict claim losses to set fair and profitable premiums.
- **Risk**
 - Wrong pricing leads to losing good customers and keeping risky
- **Challenge**
 - Highly skewed data, few costly claims, many zero claims
- **Solution Approach**
 - Apply ML models adapted for skewed, zero-heavy data



Prediction Accuracy vs. Interpretation

Accuracy

- Ensures correct premium pricing based on predicted losses
- Helps reduce adverse selection by attracting the right customers
- Directly drives business profitability and portfolio balance

Our Goal

- Ran multiple models(Stacked Random Forest, XGBoost, GLM, Two-step Approach)
- Balanced the need for high predictive power with clear and actionable insights

Interpretation

- Builds trust with business teams and regulators
- Enables transparent decision-making on premium and claims
- Identifies key risk factors influencing claim likelihood and cost



Feature Selection and Engineering

Derived Features

- days_since_first_policy_start
- days_since_last_renewal
- driver_age_days (also called age)
- driver_license_age_days
- LC
- HALC
- claims_this_year
- past_claim_only
- hp_per_kg
- value_per_hp
- premium_per_hp
- is_powerful
- policy_tenure
- policy_growth_ratio
- policy_to_driver_age_ratio
- policy_to_license_age_ratio
- claims_per_year

Interaction Terms

- Policy Tenure / Driver's Age
- Policy Tenure / Driver's License Age
- Policy Tenure $\times$ Claims Per Year
- Model Age $\times$ Claims Per Year
- Model Age $\times$ Urban
- Horsepower $\times$ Urban



Key Challenges in Predictive Modelling

Business Challenges



Imbalanced data

Many customers make *zero claims* each year → Hard to predict rare events accurately



Need for Fair and Accurate Pricing

Errors in prediction can cause **overpricing** (lose good customers) or **underpricing** (take on risky customers → financial loss)



Trade off between Accuracy and Interpretability

Complex models (like XGBoost, Stacking) give better RMSE/AUC, but are harder to explain to business leaders.

Technical Challenges



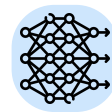
Zero-Inflated and Skewed Data

Losses have a mass at zero and a heavy right tail → Tweedie distribution models needed.



Feature Complexity and Interaction Effects

Vehicle, customer, and location features interact non-linearly → stacking and boosting help but need careful tuning



Overfitting Risk

Especially for small claimant group → managed with CV, early stopping, regularization (alpha, lambda)



Prediction and Classification Approaches

Regression Modeling: A Single-Step Approach

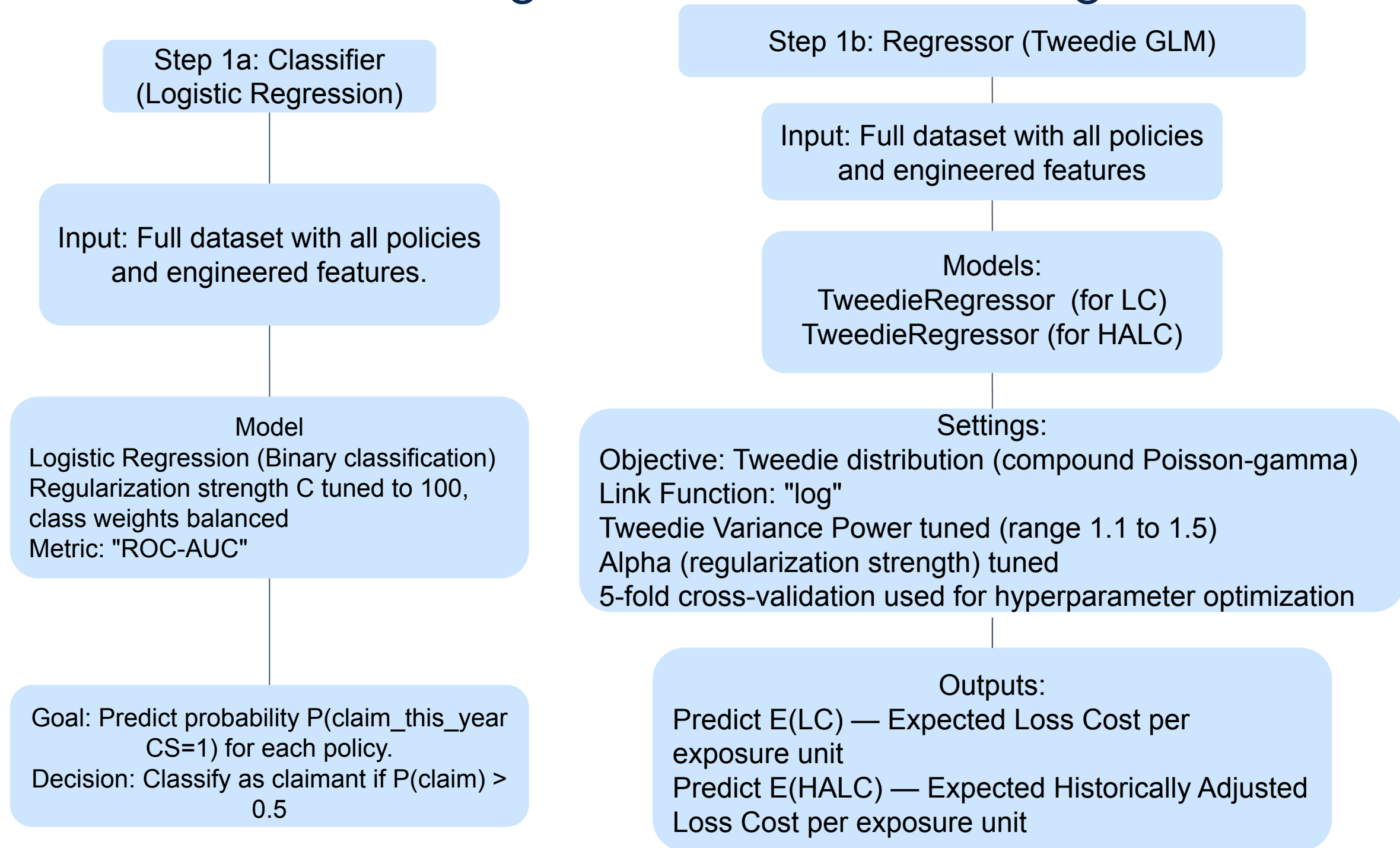
- Directly predicts **LC** and **HALC** from features without distinguishing claim status.
- Simpler pipeline: One model for LC and one model for HALC.
- Assumes all policies (whether they had claims or not) contribute similarly.
- Challenge: Handling zero-inflated and highly skewed target distributions.
- Common Models:
 - Tweedie Regressor (single step)
 - XGBoost Regressor (single step)

Classification and Regression: Two-Step Strategy

- Step 1: **Classification** to predict if a claim is likely (**claims_this_year**).
- Step 2: **Regression** only for predicted claimants to estimate **LC** and **HALC**.
- Better targets claim-specific modeling and focuses effort where claims occur.
- Addresses skewness and zero-inflation more effectively.
- Models used:
 - **Step 1:** XGBClassifier / Neural Network Classifier
 - **Step 2:** XGBRegressor / Neural Network Regressor with Tweedie Loss

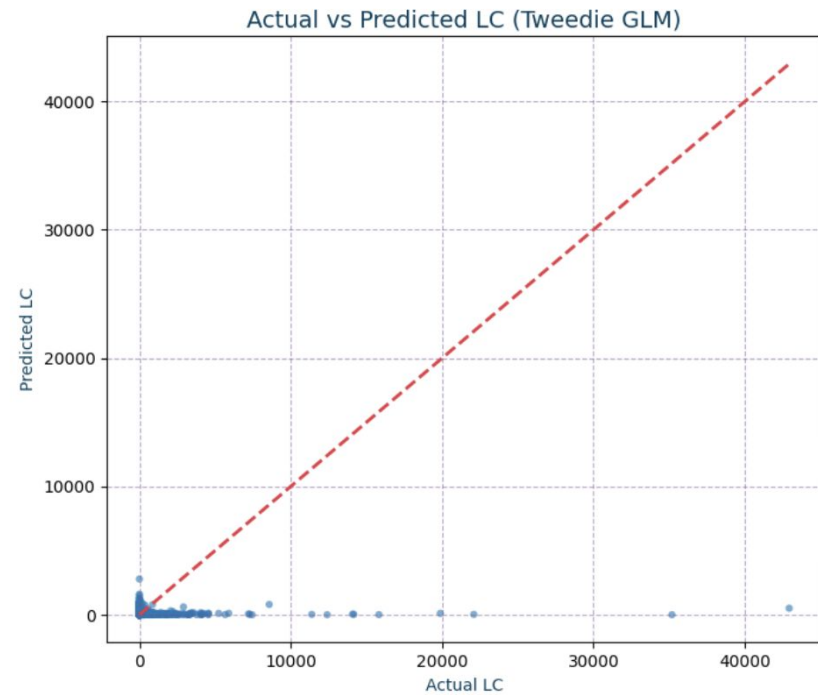
One-Step Modeling Strategy: Tweedie GLM + Logistic Regression

Challenges in Machine Learning

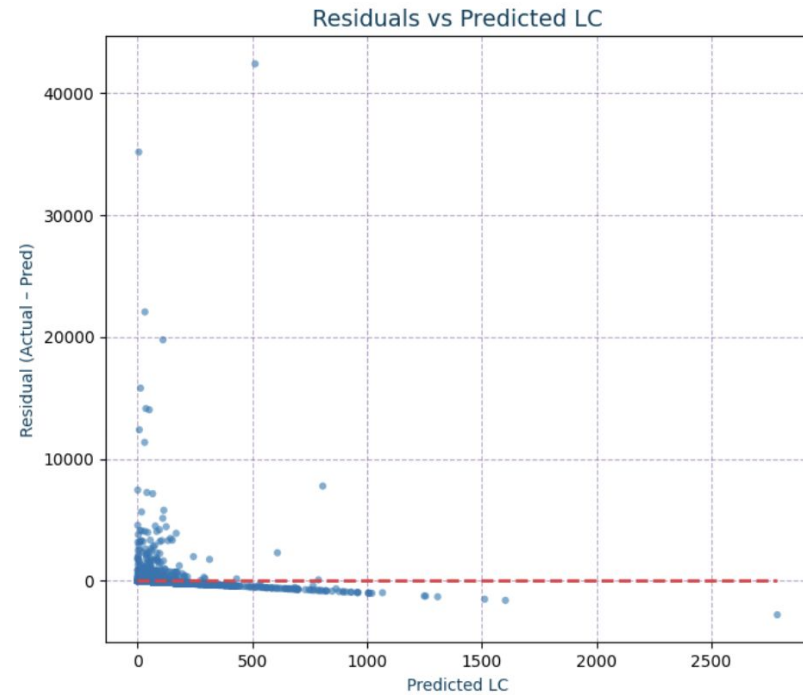


Tweedie GLM + Logistic Regression

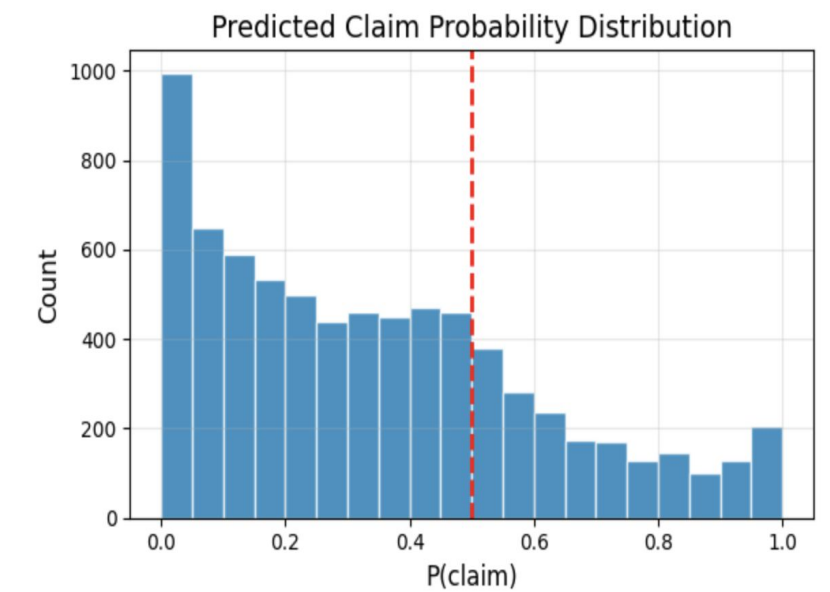
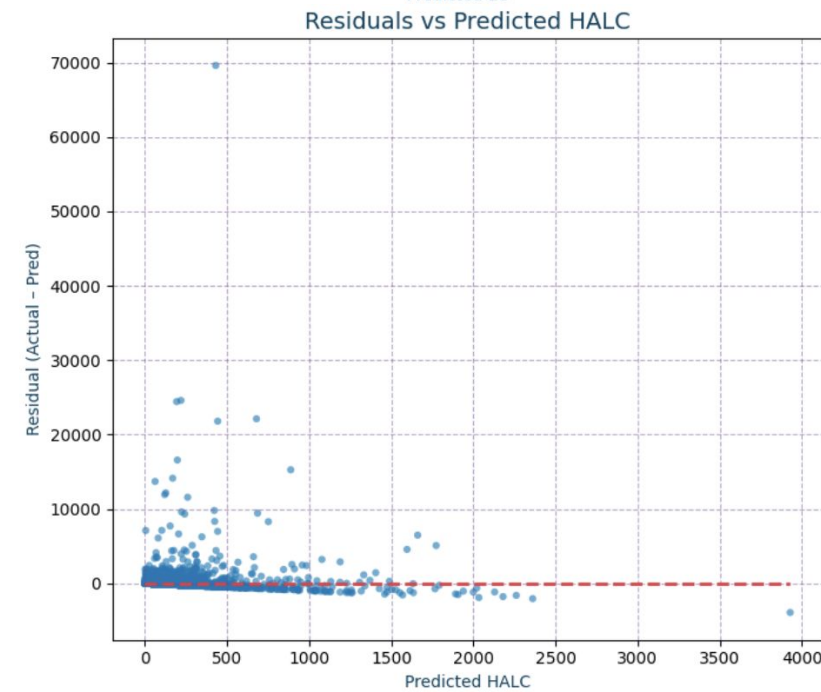
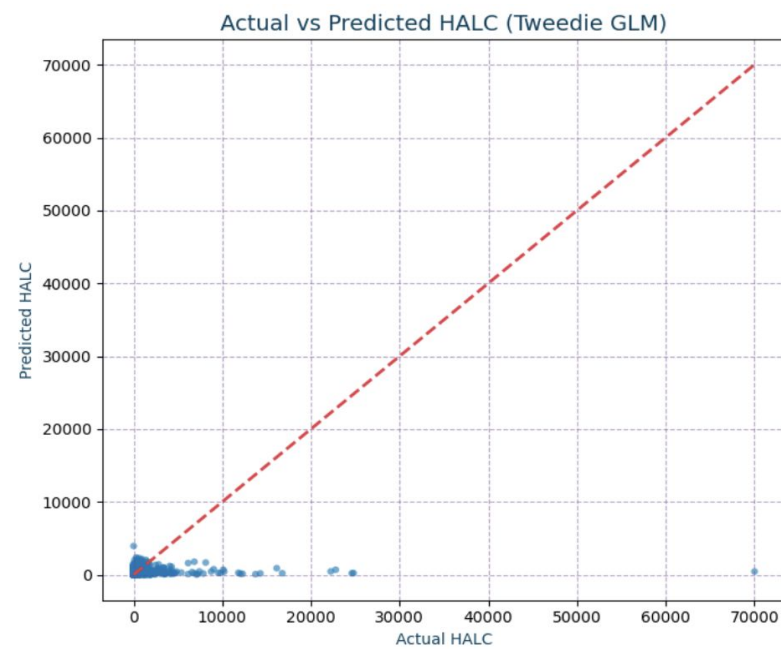
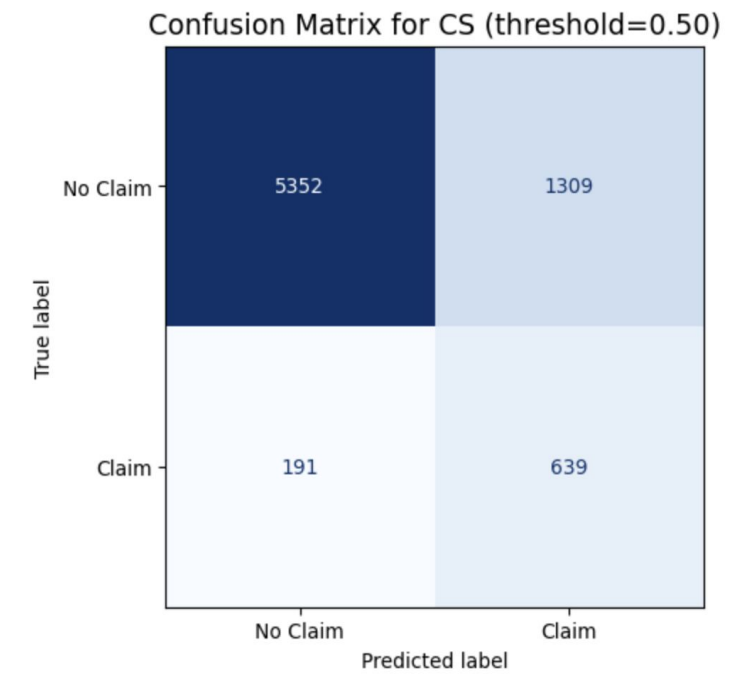
LC Evaluation



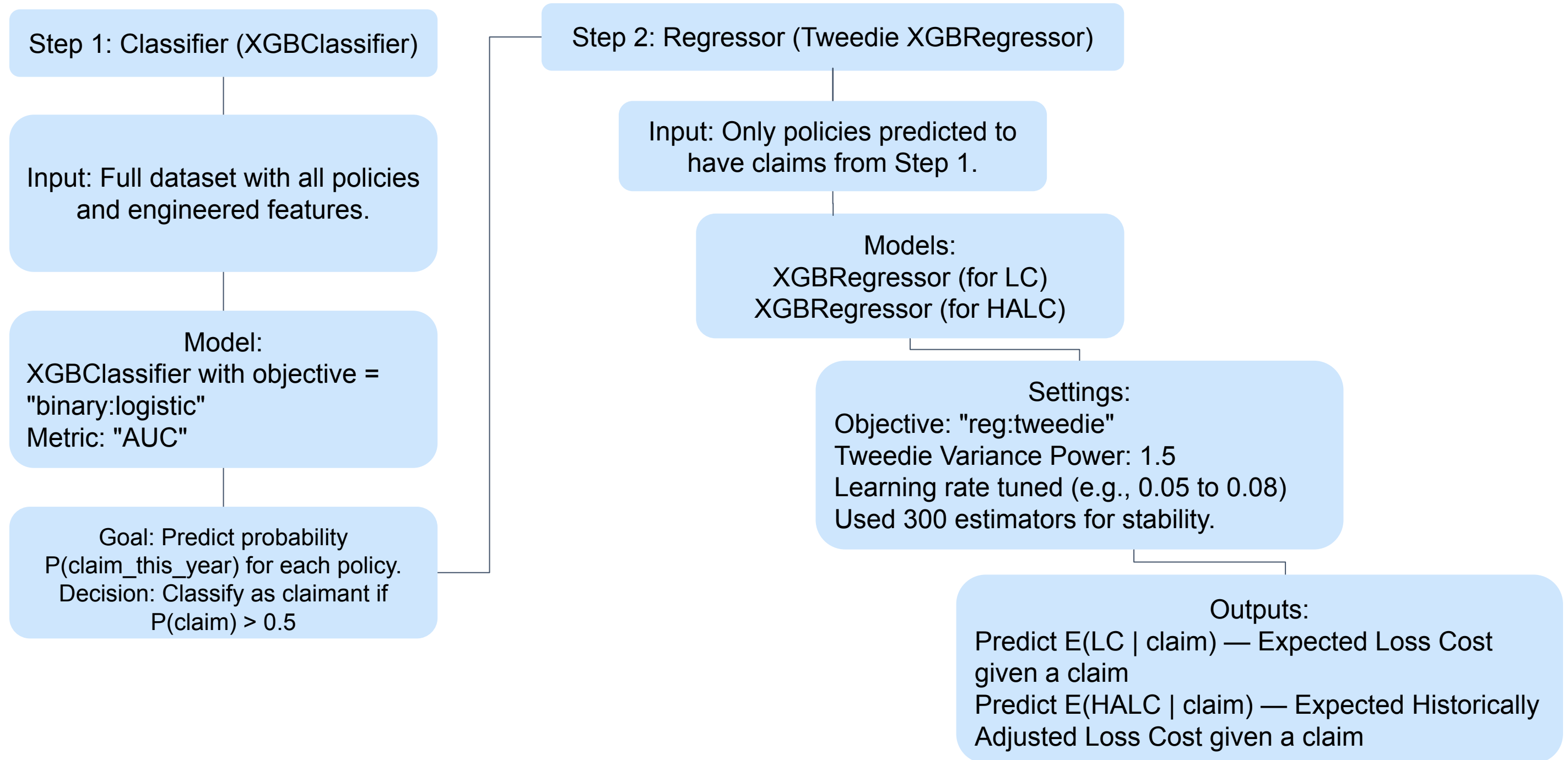
HALC Evaluation



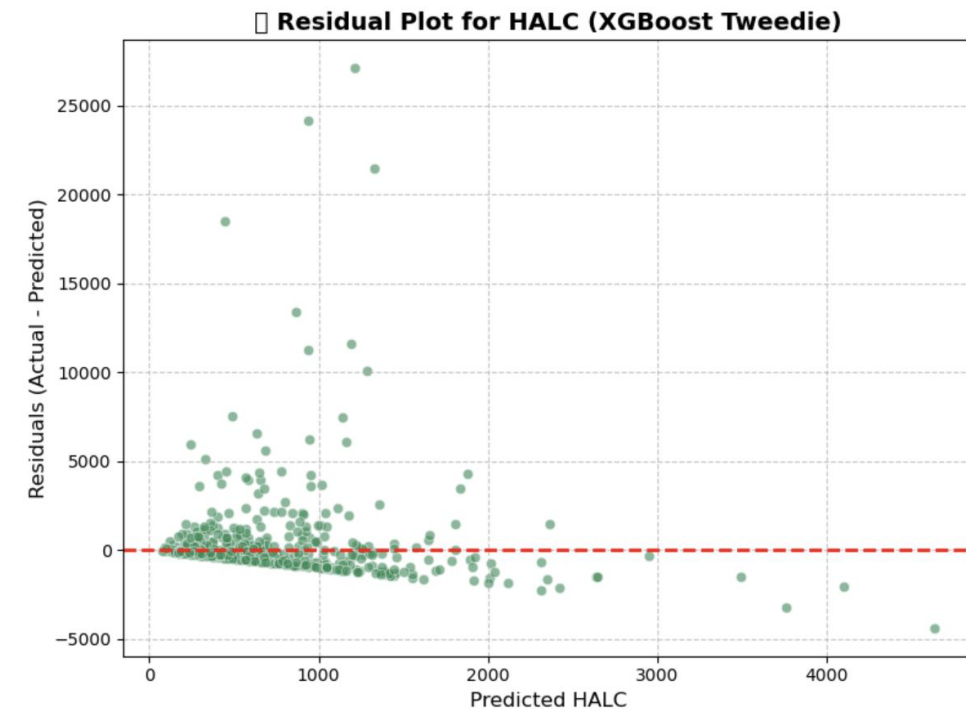
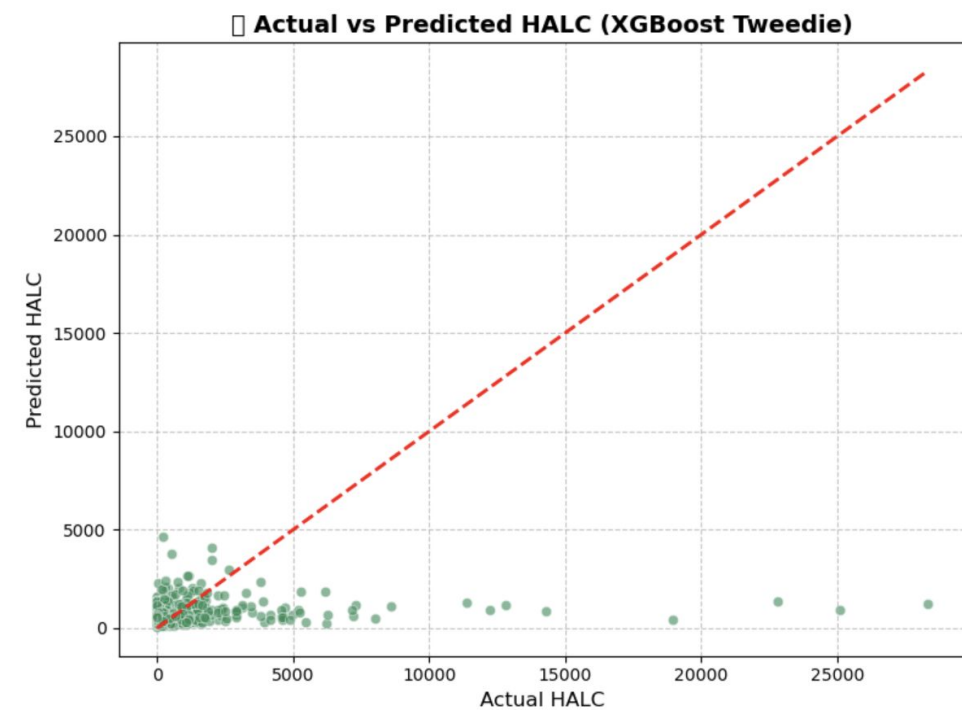
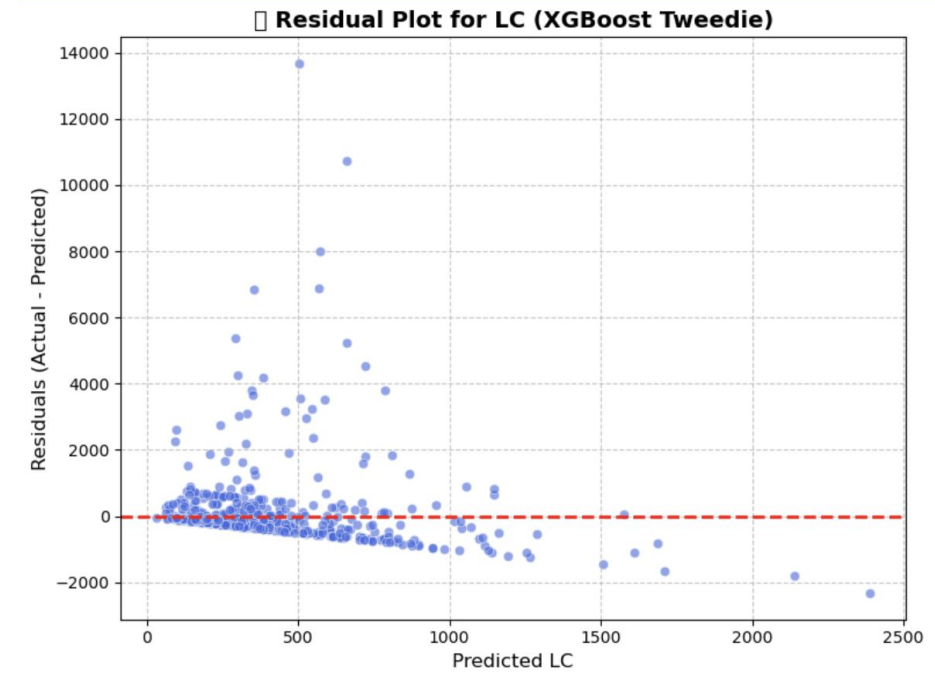
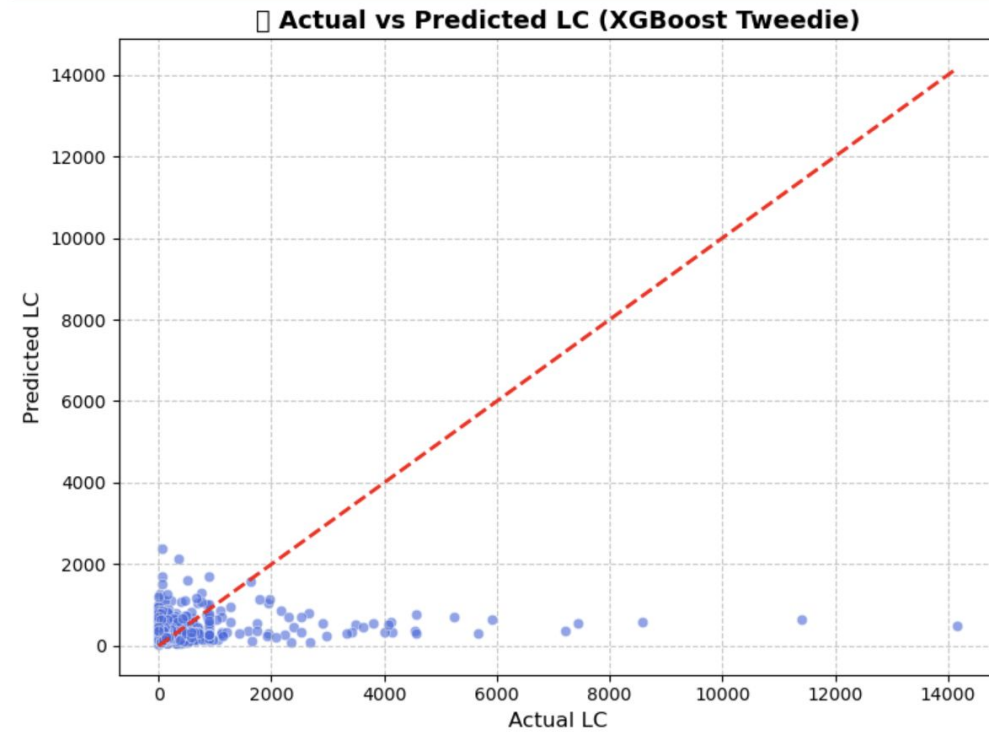
CS Evaluation



Two-Step Modeling Strategy: XGBoost Classifier + Tweedie Regressor



Two-Step Modeling Strategy: XGBoost Classifier + Tweedie Regressor



XGBoost with Tweedie

- Input with all the existing features and some feature engineering and data scaling
- Train/Test Split (80/20 on LC & HALCTarget)

Data Preparation and Preprocessing

- XGBoost Regressor (Tweedie Objective)
- Recursive Feature Elimination (RFE for Top Features)
- Hyperparameter Tuning (Optuna - 300 Trials)
- Final Model Fitting (Best Params)

LC and HALC Predictive modeling

Best trial: 249. Best value: 1425.41
Best trial: 102. Best value: 676.687

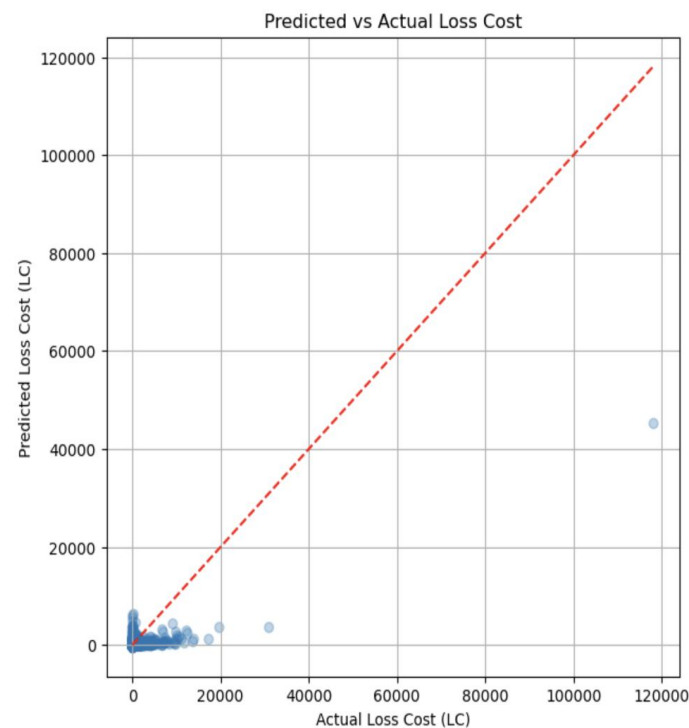
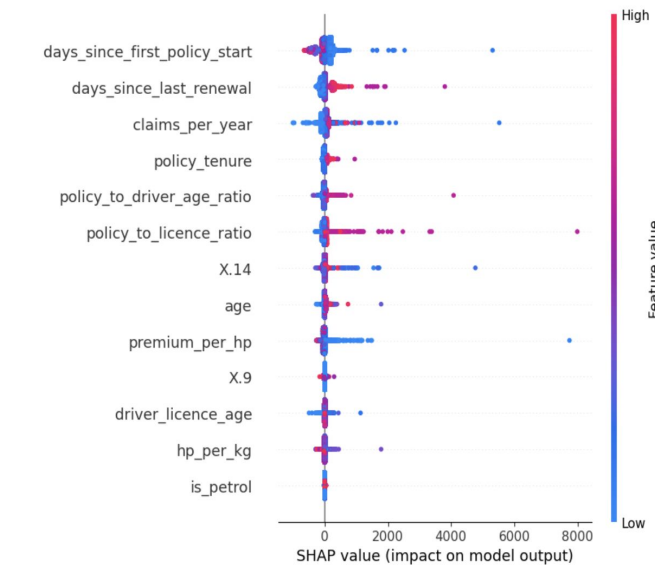
- XGBoost Classifier for Classifying CS
- Hyperparameter Tuning (Optuna - 300 Trials)
- Final best Model and predict probabilities on test

Classifier for CS

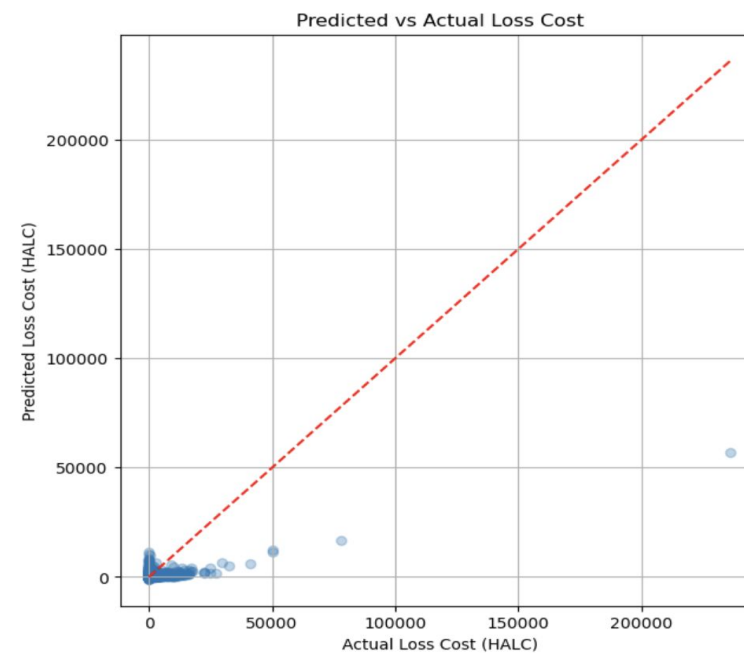
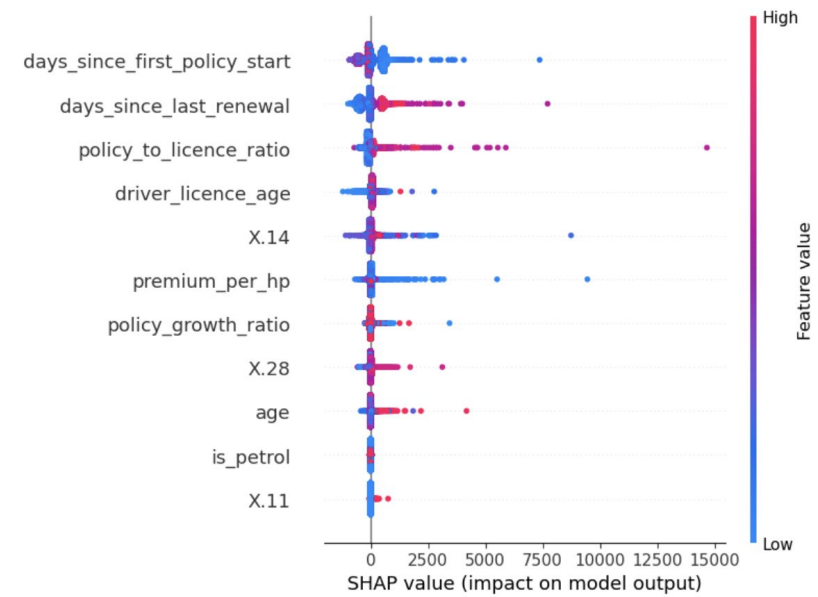
Final ROC-AUC on Test Set: 0.9491

XGBoost with Tweedie

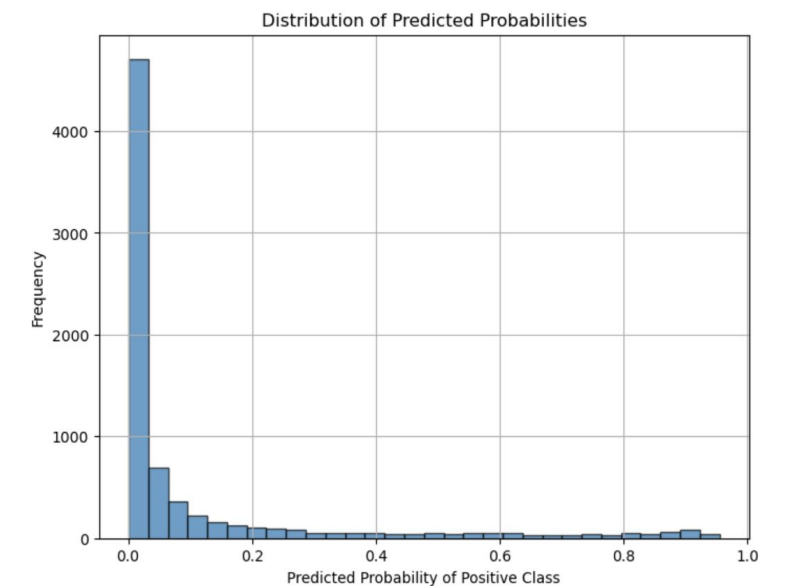
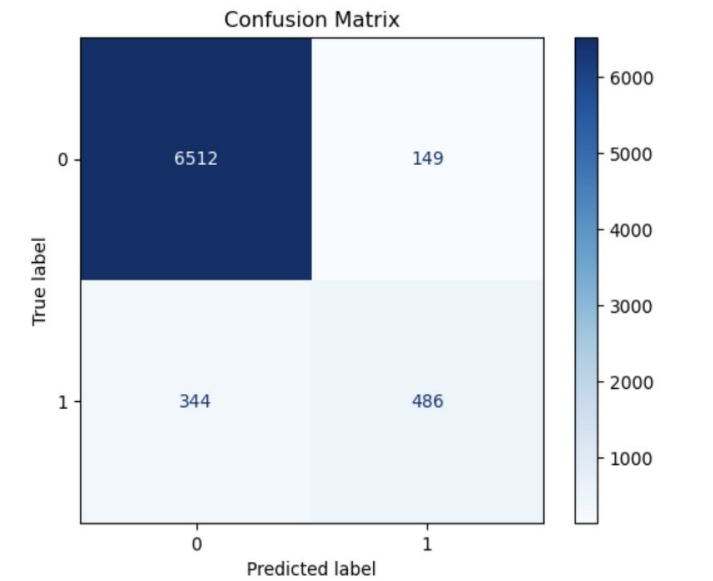
LC Evaluation



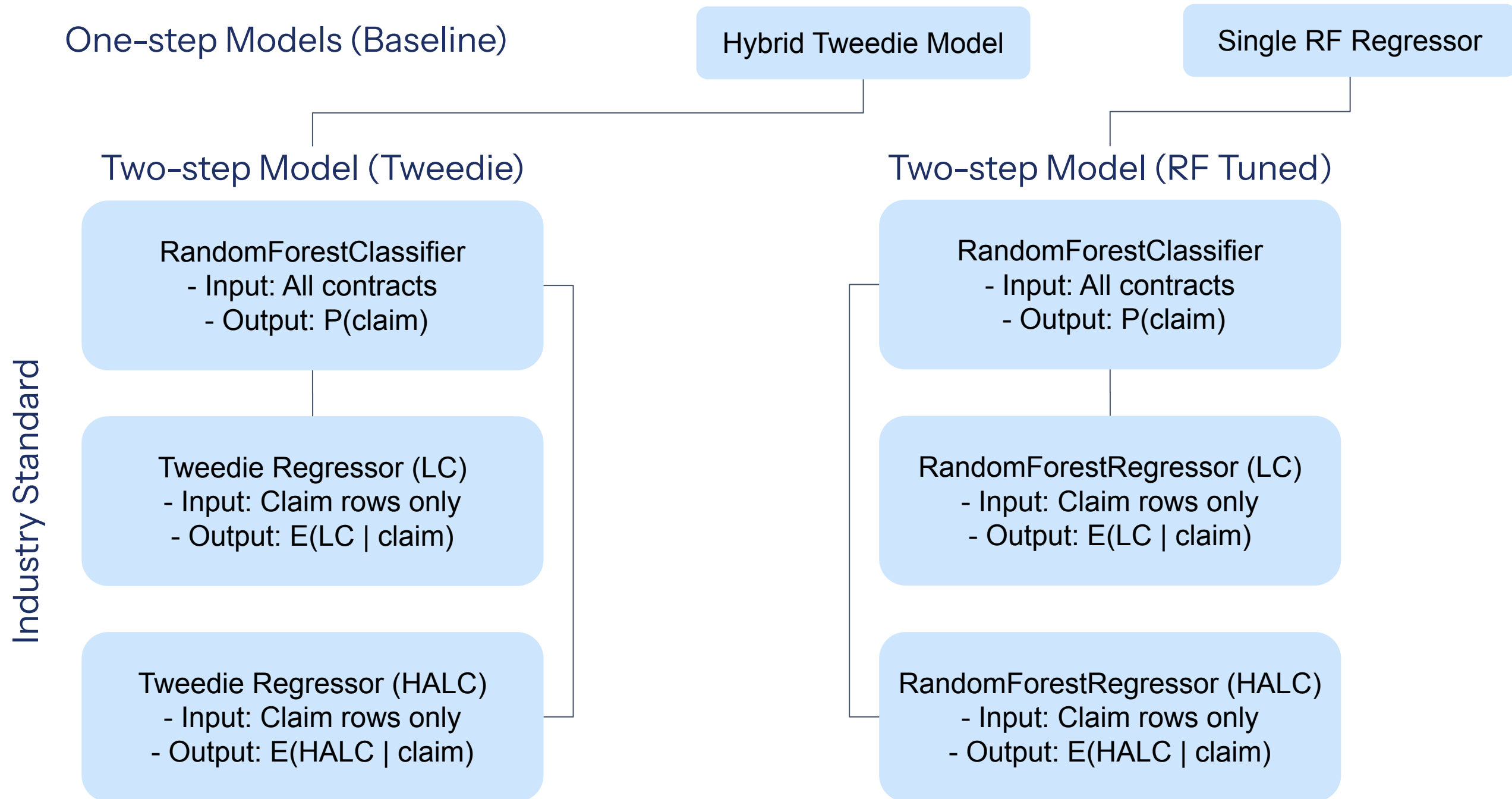
HALC Evaluation



CS Evaluation

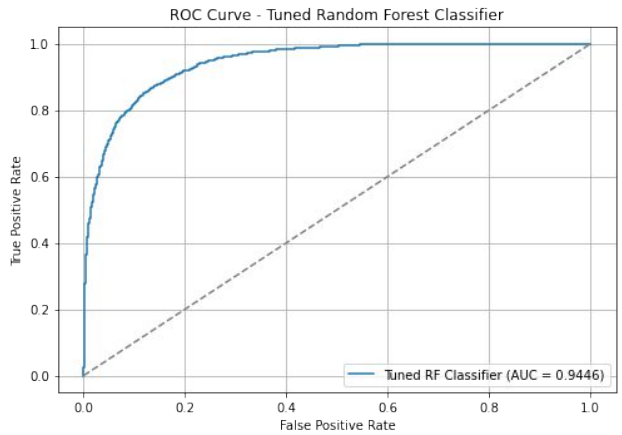
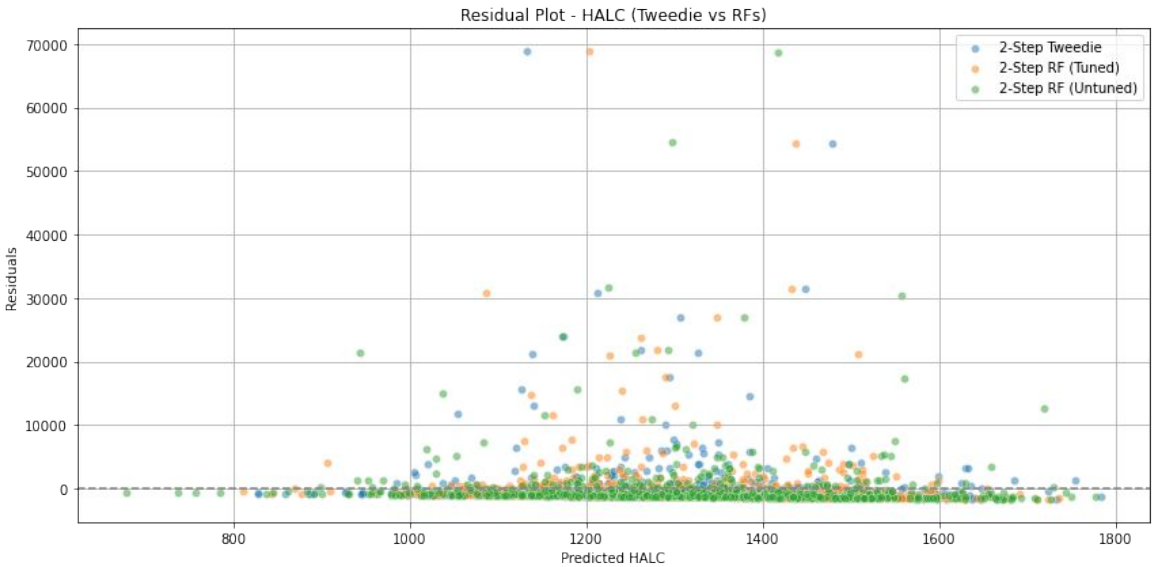
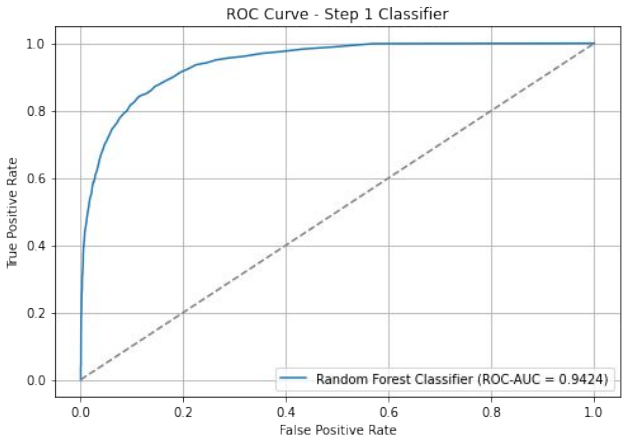
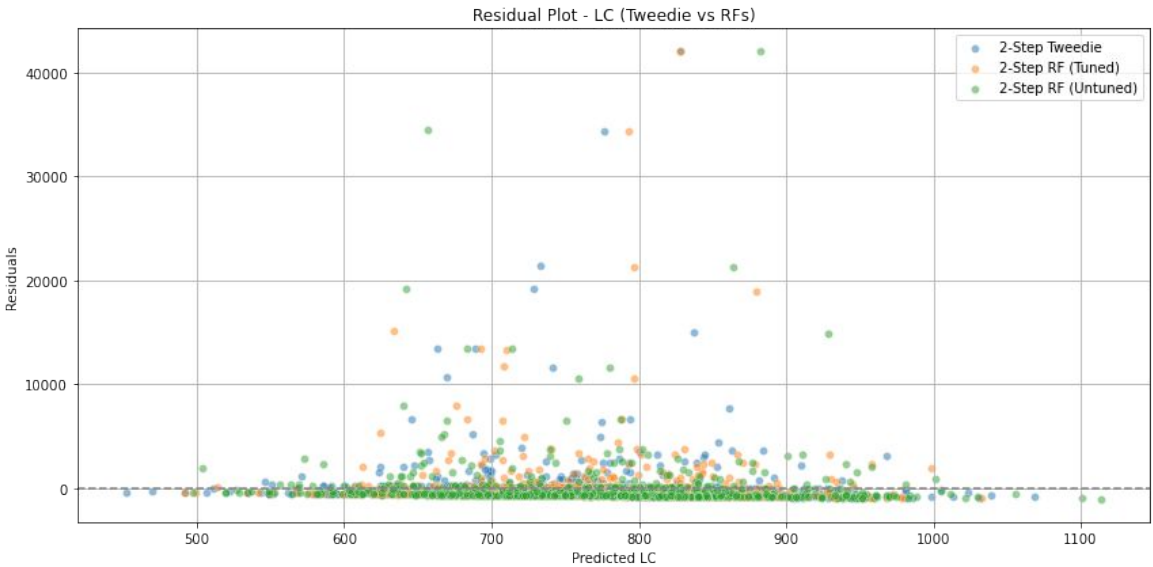


Random Forest and Tweedie Regressor Models



Random Forest and Tweedie Regressor Models

MODEL TYPE	RMSE LC	RMSE HALC	ROC-AUC	NOTES
HYBRID TWEEDIE	2,091.5	1,495.0	-	✗
HYBRID RF	1,039.63	1,703.76	-	✗
HYBRID RF (TUNED)	870.13	1,457.39	-	VAST IMPROVEMENT
2-STEP TWEEDIE	868.89	1,457.25	0.94	✓ Industry Standard
2-STEP RF (TUNED)	868.56	1,454.02	0.94	✓ BEST PERFORMANCE

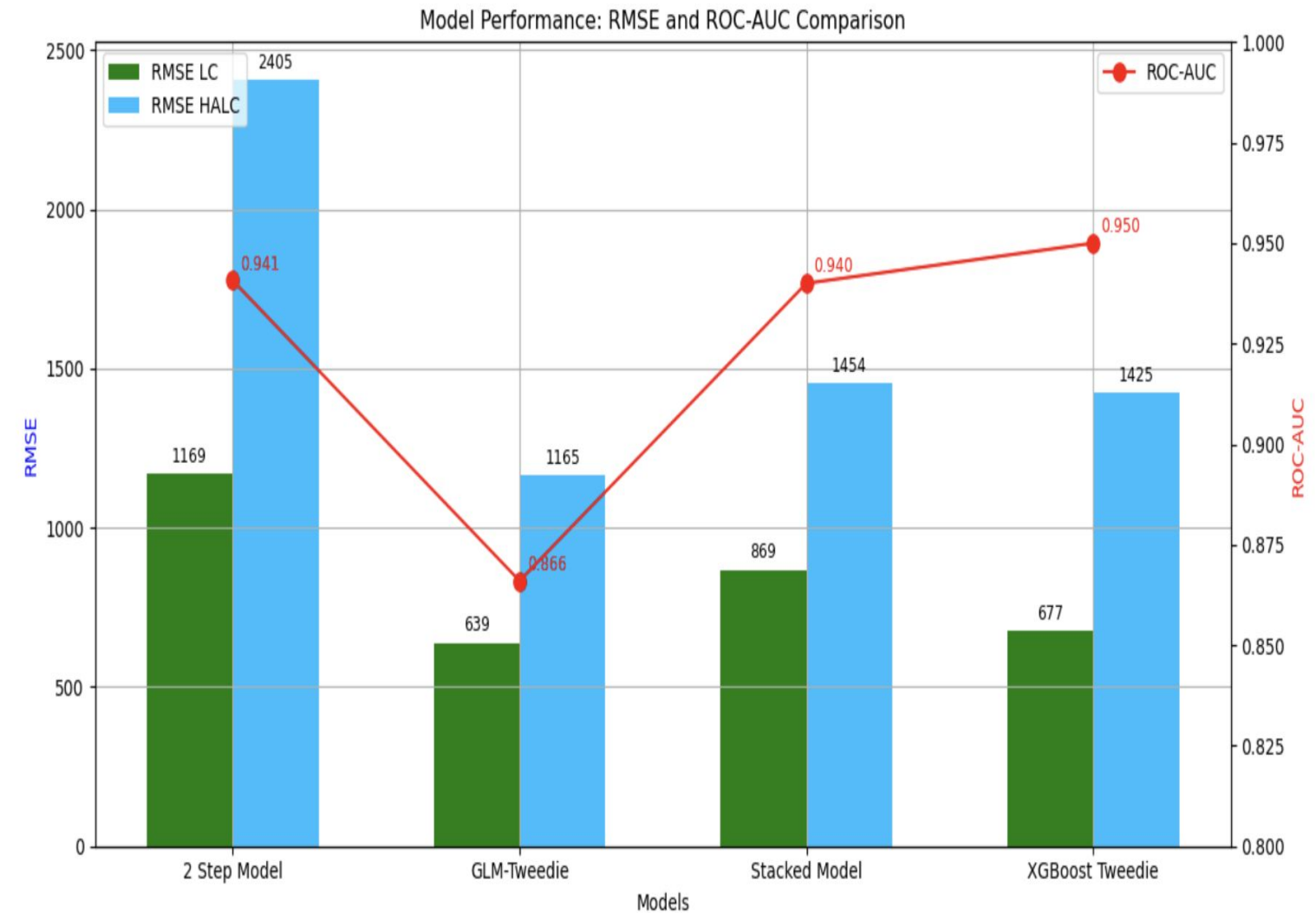


Model Comparison

Key Observations:

- XGBoost Tweedie achieves the lowest RMSE for both LC (676) and HALC (1425), and highest ROC-AUC (0.95).
- GLM-Tweedie provides better interpretability but lower predictive performance (especially ROC-AUC ~ 0.866).
- Stacked Model is slightly worse than XGBoost Tweedie on RMSE and ROC-AUC — indicating marginal improvements from stacking, but not significant.

Model	Interpretability	Predictive Accuracy
GLM-Tweedie	High	Moderate
Stacked Model	Moderate	Good
XGBoost Tweedie	Low	Best

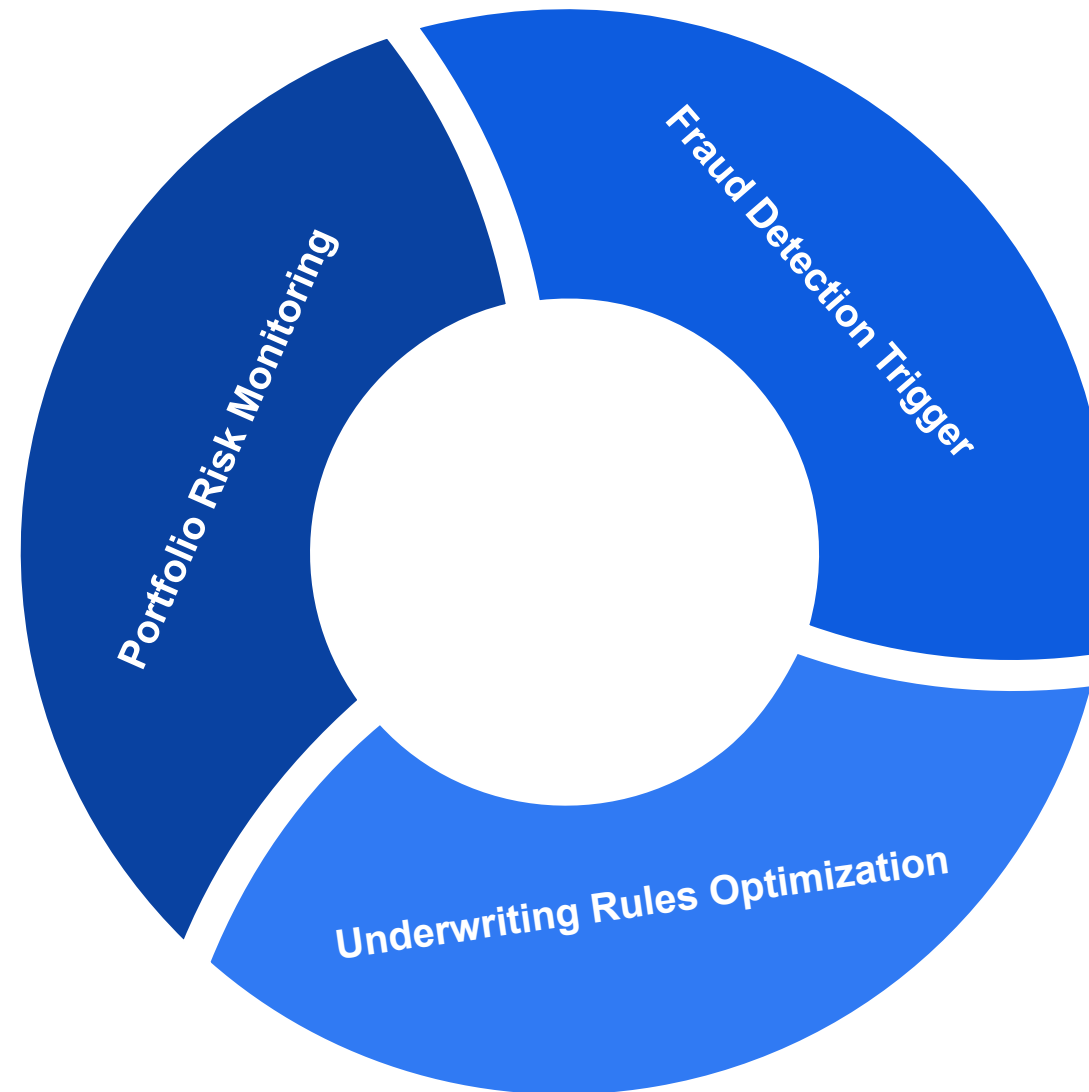


Future Directions

Aggregate individual predictions into book-level risk metrics.

How:

- Compute total expected loss
- Forecast total claims
- Build dashboards by segmenting vehicle type, urban vs rural, distribution channel



Flag policies where predicted Loss Cost per Exposure Unit (LC) or claim probability (CS) are extreme outliers.

How:

- Define upper-tail thresholds on LC (e.g. 99th percentile of LC).
- Cross-filter to “low-risk” segments, e.g.:
Young drivers (age < 25 years)

Refine accept/reject rules using model-derived feature importance

How:

- Rank features by SHAP or coefficient magnitude,
- Use partial-dependence plots on those features to find risk “breakpoints.”
- Tighten rules for policies

Why Life Insurance needs its own modeling approach



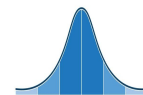
Different Risk Dynamics

- **Auto:** Losses driven by accidents, mileage, vehicle age/attributes
- **Life:** Payouts governed by mortality/morbidity and survival timelines



Mismatched Data Features

Auto Inputs	Life Inputs
Vehicle horsepower, age, urban vs. rural location, past claim counts, policy tenure	Age, Gender, Health history, lifestyle,, biometric readings, demographics



Statistical Distribution Differences

- **Auto claims:** “Zero-inflated” → many policies have \$0 loss, then a heavy right tail (Tweedie)
- **Life payouts:** Time-to-event → everyone eventually dies after >0 time; no spike at zero



Regulatory & Ethical Constraints

- **Life pricing:** Governed by prescribed mortality tables and anti-discrimination laws
- **Actuarial standards:** Requires specialized models, expert oversight, and strict compliance

